

Beyond the Rate: Information Content of FOMC Statement Language

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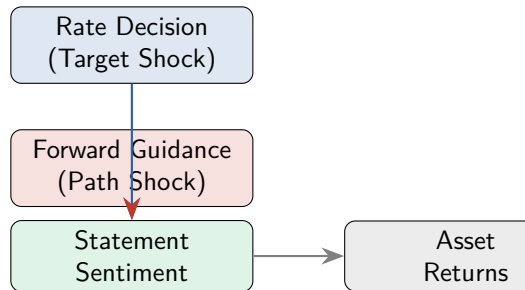
The Puzzle

Central question:

Does FOMC statement language convey information **beyond** the rate decision?

Why it matters:

- Central banks increasingly rely on **forward guidance** as a policy tool
- Markets respond to *what the Fed says*, not just *what it does*
- The “information channel” (Romer & Romer, 2000) suggests FOMC statements reveal the Fed’s superior economic assessment



- **Kuttner (2001)**: 30-min FF futures window → isolate monetary policy surprises
- **GSS (2005a)**: Decompose surprises into **target** + **path** factors
- **Nakamura & Steinsson (2018)**: High-frequency identification of monetary non-neutrality
- **Campbell et al. (2012)**: Delphic vs. Odyssean forward guidance
- **Jarociński & Karadi (2020)**: Separate policy shock from information shock
- **Hansen et al. (2018)**: Computational linguistics for FOMC transparency
- **New (2024–25)**: Chen et al. (2025) GPT-4 topic decomposition; Gambacorta et al. (2024) CB-LMs; Yao & Chai (2025) uncertainty-aware LLM

Our contribution:

- 1 First to directly link **GSS target/path shocks** to **FOMC statement sentiment**
- 2 Expanded CB dictionary (591 hawkish + 222 dovish terms, fully disjoint)
- 3 Formal test of the **information channel** via text analysis
- 4 **New**: Dual-equation test, forward-lookingness decomposition, novelty weighting

Monetary Policy Shocks

- Acosta (2022): 220 FOMC meetings, 1995–2022
- **Target shock**: surprise in current FF rate
- **Path shock**: surprise in future rate path
- Extension: USMPD (SF Fed) → 2026

Market Data

- CRSP VW/EW/S&P 500 via WRDS
- 117 meetings with complete overlap

FOMC Statement Sentiment

- 164 statements, 2006–2025
- Expanded CB dictionary:
 - 591 hawkish terms
 - 222 dovish terms
 - Fully disjoint
- Score = $\frac{H-D}{H+D+\epsilon}$
- Validated: $r = 0.29$ with rate change
- **New**: CB-only outperforms combined (R^2 : 3.90% vs 1.57%)

Four hypotheses:

Hypothesis	Prediction	Specification
H1	Shocks predict sentiment	$S_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H2	Shocks predict returns	$R_t = \alpha + \beta_1 \text{Target}_t + \beta_2 \text{Path}_t + \varepsilon_t$
H3	Path > Target for sentiment	$ \beta_2 > \beta_1 $ in H1
H4	Stronger during FG period	Interaction with FG dummy

Estimation: OLS with Newey-West (lag=4, data-driven) standard errors

Sample: 117 FOMC meetings, January 2006 – July 2022

New experiments:

- 1 Dual-equation regression (equity + risk premium channels)
- 2 Forward-lookingness decomposition (FL vs. CA sentiment)
- 3 Statement novelty weighting (WLS)

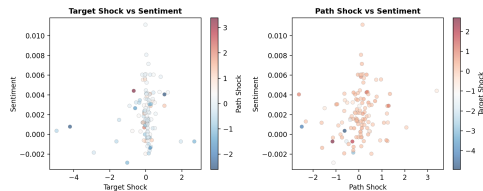
H1: Sentiment and Monetary Policy Shocks

	Coefficient	p -value
Target shock	0.000577	0.017**
Path shock	0.000633	0.152
R^2	1.57%	
N	117	

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Key finding: Target shock is significant at 5%, path is **not**.

Wald test: $\chi^2 = 0.02$, $p = 0.90$ — cannot reject $\beta_1 = \beta_2$.



Sentiment Measure Comparison

Measure	R^2	$p(\text{Target})$	$p(\text{Path})$
CB-only	3.90%	$<0.05^{**}$	0.031^{**}
Combined ($0.5 \times \text{LM} + 0.5 \times \text{CB}$)	1.57%	0.017^{**}	0.152
LM-only	0.33%	0.474	0.552

Key insight:

- CB-only **doubles** R^2 and makes path shock significant at 5%
- LM component adds **noise without signal** for FOMC text (always positive)
- Domain-specific dictionaries outperform general-purpose financial dictionaries

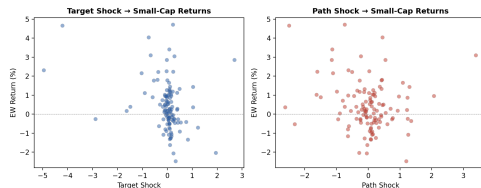
H2: Asset Returns and Shocks

Asset	β_T	p_T	β_P	p_P
CRSP VW	-0.435	0.043**	-0.186	0.434
CRSP EW	-0.449	0.013**	-0.174	0.421
S&P 500	-0.391	0.073*	-0.179	0.426
Gold	-0.404	0.016**	0.021	0.935

Returns in percentage points; $N = 117$

Findings:

- Target shock: significant for EW & Gold
- Path shock: **not significant** for any asset
- Small-cap sensitivity consistent with literature



H3 & H4: Information Channel and Forward Guidance

H3 — Path > Target?

- Target significant at 5%, path is not
- Wald test: $p = 0.90$ — cannot reject equality
- Evidence is *suggestive*, not definitive

H4 — Stronger during FG period?

- Sentiment $\times$ FG interaction: $p = 0.602$ (CRSP VW), $p = 0.041$ (NASDAQ, but economically implausible)
- **Not robustly significant** — language channel does NOT strengthen at ZLB
- Three interpretations:
 - ① Shocks absorb information regardless of regime
 - ② Dictionary cannot distinguish FG language (cf. GPT-3.5 fails 135/139 statements)
 - ③ FG operates through risk premium channel (Chen et al. 2025)
- **Key finding:** Rate cut regime — path shock **highly significant** ($p < 0.001$, $R^2 = 43.1\%$)

Experiment 1: Dual-Equation Regression

Hypothesis: Forward guidance operates through a risk premium channel (Chen et al. 2025)

Dependent Variable	R^2	$p(\text{Target})$	$p(\text{Path})$
<i>Channel 1: Equity Returns (Expectations)</i>			
CRSP VW	9.1%	0.040**	0.440
S&P 500	7.8%	0.072*	0.426
<i>Channel 2: Risk Premium (Bond Market)</i>			
10Y Treasury chg	0.72%	0.403	0.890
VIX change	0.24%	0.970	0.401
Term spread chg	1.39%	0.667	0.341

Result: Risk premium channel **not detectable** at daily frequency.

Consistent with Chen et al. using 30-min windows — channel too fast for daily data.

Experiment 2: Forward-Lookingness Decomposition

Hypothesis: Path shock captures forward-looking language (dimension mismatch)

Sentiment Measure	R^2	$p(\text{Target})$	$p(\text{Path})$
Combined (baseline)	4.12%	0.099	0.015**
Forward-looking only	0.79%	0.012**	0.800
Current-assessment only	1.30%	0.012**	0.506

Counterintuitive finding:

- Path shock is **most significant** in the combined measure ($p = 0.015$)
- Path shock is **least significant** in forward-looking sentiment ($p = 0.80$)
- **Dimension mismatch hypothesis rejected**
- Path shock captures *broad policy stance signal*, not specifically forward guidance
- GSS target/path decomposition $\neq$ FOMC text FL/CA dimensions

Experiment 3: Statement Novelty Weighting

Idea: Not all FOMC statements are equally informative

Method	R^2	Improvement
Unweighted OLS	3.98%	baseline
Novelty-weighted WLS	5.75%	+45%

Novelty = cosine distance between TF-IDF vectors of consecutive statements

Subsample analysis:

- High novelty statements: $R^2 = 5.78\%$
- Low novelty statements: $R^2 = 6.53\%$
- Difference is small — novelty helps via weighting, not selection

Absolute R^2 remains modest — measurement innovation alone cannot compensate for daily-frequency limitation.

Robustness Checks

Specification	R^2	$p(\text{Target})$	$p(\text{Path})$	N	Period
Full (Acosta)	1.57%	0.017**	0.152	117	2006–22
CB-only	3.90%	$<0.05^{**}$	0.031**	117	2006–22
No COVID	1.57%	0.017**	0.154	115	2006–22
Post-2010	0.59%	0.117	0.258	97	2010–22
Rate change proxy	0.40%	0.323	0.323	117	2006–22
Extended (USMPD)	1.65%	0.419	0.058*	163	2006–26

Key takeaways:

- CB-only sentiment: best specification ($R^2 = 3.90\%$, both shocks significant)
- Rate change proxy: $4\times$ lower R^2 — identification matters
- Extended sample: path shock significant at 10% through 2026

What the New Literature Tells Us

Chen et al. (2025): GPT-4 Topics

- Forward guidance operates through **risk premium channel**
- GPT-3.5 fails to identify FG in 135/139 statements
- Our CB dictionary has the same limitation

Gambacorta et al. (2024): CB-LMs

- Open-weight, domain-specific models
- Outperform dictionaries on stance classification
- **Most promising upgrade path**

IMF (2025): Four Dimensions

- Topic + Stance + Sentiment + Forward-lookingness
- Topic-level sentiment > aggregate
- Our null H4 may reflect aggregation

Yao & Chai (2025): Uncertainty

- LLM confidence scores
- Downweight ambiguous classifications
- Could improve signal-to-noise ratio

Limitations:

- ① **Low R^2 :** Shocks explain $<4\%$ of sentiment variation
- ② **Dictionary-based sentiment:** Cannot capture context or pragmatics
- ③ **Daily frequency:** Too coarse to detect risk premium channel
- ④ **No JK decomposition:** Cannot separate MP vs. information shocks
- ⑤ **Sample size:** $N = 117$ limits sub-sample power

Priority upgrades:

- **CB-LM embeddings** (Gambacorta et al. 2024) — reproducible + contextual
- **High-frequency data** (30-min windows) — detect risk premium channel
- **JK sign restrictions** — separate MP vs. information shocks
- Uncertainty-aware classification (Yao & Chai 2025)
- Cross-country: ECB, BoJ, BoE

Conclusion

- 1 The **target shock** is a statistically significant predictor of FOMC statement sentiment ($p = 0.017$)
- 2 The **path shock** is **not** significant ($p = 0.152$), contradicting the strong information channel
- 3 **CB-only sentiment** doubles R^2 and makes path significant — domain-specific measurement matters
- 4 The forward guidance interaction is **completely null** — language channel does not strengthen at ZLB
- 5 **New:** Risk premium channel not detectable at daily frequency
- 6 **New:** Path shock captures broad policy stance, not specifically forward guidance
- 7 **New:** Novelty weighting improves R^2 by 45%

Takeaway: FOMC language is informative, but monetary policy shocks are just one of many forces shaping it. Understanding the full information content requires looking beyond the rate decision — and beyond the shocks themselves.

Thank You

Questions & Discussion

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Appendix: Event Study Methodology

Market Model (Brown & Warner, 1985):

- ① Estimation: $R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it}$
- ② Abnormal Return: $AR_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{mt})$
- ③ CAR: $CAR_i = \sum_{t=-T_1}^{T_2} AR_{it}$
- ④ t-stat: $t = \frac{CAR_i}{\sigma_{AR} \times \sqrt{N}}$

Important: The denominator is $\sigma_{AR} \times \sqrt{N}$, **not** $\sigma_{AR}/\sqrt{N}$.

- σ_{AR} : std. dev. of abnormal returns from estimation window
- N : number of days in event window
- Common error: using $\sigma_{AR}/\sqrt{N}$ inflates t by factor of N

Appendix: Literature Radar

Automated daily scan of SSRN, arXiv, BIS/IMF/Fed, top journals

Tier	Count
High relevance (≥ 0.4)	2
Medium relevance (0.3–0.4)	11
Low relevance (< 0.3)	29
Already cited	4

Key papers discovered:

- Chen, Granville & Matousek (2025) — PLMIS shocks from GPT-4
- Gambacorta et al. (2024) — Central Bank Language Models
- IMF WP 2025/109 — Four-dimensional communication analysis
- Yao & Chai (2025) — Uncertainty-aware LLM classification
- Weinig (2025) — Multi-agent narrative MP surprises